Structured Products
2026

Johann BARCHECHATH

1. Case Studies

1. http://structuredproducts.free.fr

- Top right : you can change FR ⇔ EN
- Login / password given in course
- Forbidden cases : 1-5, 11, 21 (and all case seen during the course)
- Number of stars = indication of difficulty level
- The lower the difficulty, the highest the exigence for quality in pricing and presentation

2. Pick up your case

- The delegate is in charge to feed "Choice of Case Study.xls" on "first come, first served" basis
- In "Choice of Case Study.xls", you will find the max number of student per case (no name in gray zone => ask for exceptions)
- 1 case study for 1 group (<u>bijection</u>)
- Case 16 : a lot of fun … with matrix of correlation

© J. BARCHECHATH

1. Case Studies

3. Work to be done

- As per detailed on the website & the case :
  - PowerPoint (max size 3Mo)
  - Excel pricer (use the template)

- The Excel is provided in the website WITH the market data you have to use:
  - Yield curve sheet
  - Vol surface sheet (include the Spot)

- Is **your responsibility** to come to me such that I validate your understanding of the case, payoff, composition of the product. If I don't validate it, you have more than 50% of "chance" to fail the exam.

- To be send via K2 by 10th of April 2h before the presentation.
- Case Study presentation the 10th / 11th of April (TBD)

© J. BARCHECHATH

1. Case Studies

3. More information

- There is no more "Bourse des questions" (getting points for answering other team questions)
- No respect of any rule = penalties up to 5 points per infraction
- Excel file should NOT be renamed
- All files should include all team members names

- In ANY email, precise your class & campus !
- In ANY email, precise the FULL name of the case, and all the students in the group.

© J. BARCHECHATH

2. Useful formulas

1. B&S assumption

$$\ln S_T - \ln S_0 \sim N\left(\left(\mu - \frac{\sigma^2}{2}\right)T, \sigma\sqrt{T}\right)$$

by discretisation for positive value

$$\ln \frac{S_T}{S_0} = \left(\mu - \frac{\sigma^2}{2}\right) \times T + \sigma\sqrt{T} \times \varepsilon \quad \textit{with } \varepsilon \sim N(0;1)$$

$$S_T = S_0 \times \exp\left(\left(\mu - \frac{\sigma^2}{2}\right) \times T + \sigma\sqrt{T} \times \varepsilon\right)$$

© J. BARCHECHATH

2. B&S (no div)

$$Call = S_0\, N(d_1) - K\, e^{-rT} N(d_2)$$

where $$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

$$Digital\ Call = e^{-rT} N(d_2)$$

r = continuous risk free rate

© J. BARCHECHATH

3. B&S (div or FX)

- Equity : r = risk free and q = dividend rate
- FX : r = $ (ccy2) rate and q = € (ccy1)rate

$$Call = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2)$$

$$Put = K e^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1)$$

where $$d_1 = \frac{\ln(S_0 / K) + (r - q + \sigma^2 / 2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0 / K) + (r - q - \sigma^2 / 2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

© J. BARCHECHATH

4. Black (for Fixed Income)

The value of a caplet, for period $[t_k, t_{k+1}]$ is :

$$L\delta_k P(0,t_{k+1})[F_k N(d_1) - R_X N(d_2)]$$

where $d_1 = \dfrac{\ln(F_k / R_X) + \sigma_k^2 t_k / 2}{\sigma_k \sqrt{t_k}}$ and $d_2 = d_1 - \sigma\sqrt{t_k}$

F_k : forward interest rate for (t_k, t_{k+1})
σ_k : interest rate volatility
L : principal
R_X : cap rate
$\delta_k = t_{k+1} - t_k$
$P(0, t_{k+1}) = exp(-r(t_{k+1})*t_{k+1})$

© J. BARCHECHATH

5. How to smile in FX ?

$$RR(25\Delta) = \sigma_C(25\Delta) - \sigma_P(25\Delta)$$

$$BF(25\Delta) = \frac{\sigma_C(25\Delta) + \sigma_P(25\Delta)}{2} - \sigma_{ATM}$$

$$\sigma_C(25\Delta) = \sigma_{ATM} + BF(25\Delta) + 0.5 \cdot RR(25\Delta)$$

$$\sigma_P(25\Delta) = \sigma_{ATM} + BF(25\Delta) - 0.5 \cdot RR(25\Delta)$$

6. How to interpolate vol ?

- Between 2 strikes : linear
- Between 2 dates :

$$\sigma(t) = \sqrt{\frac{1}{t} \cdot \left[t_1\sigma_1^2 + \frac{(t - t_1)}{(t_2 - t_1)} \cdot \left(t_2\sigma_2^2 - t_1\sigma_1^2\right) \right]}$$

© J. BARCHECHATH

7. Interest rates

- Bases : Act/360, 4% p.a., …
- Prorata, actuarial, ^m.T, continuous
- If Rate is compounded then :

$$DF_T = \frac{1 - \frac{Rate_T}{Rate_{T-1}} \times (1 - DF_{T-1})}{1 + Rate_T}$$

$$DF_T = \frac{1}{(1 + ZC_T)^T}$$

$$DF_T = e^{-T \times ZC_T}$$

$$ZC_t = ZC_{t1} \times \frac{t_2 - t}{t_2 - t_1} + ZC_{t2} \times \frac{t - t_2}{t_2 - t_1}$$

© J. BARCHECHATH